STUDENT SKILLS DEVELOPMENT MANAGEMENT SYSTEM

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

CSA4305- Internet Programming

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

T Gnaneswari(192372183)

P Madhavi(192372133)

M Ramya sri(192372144)

V Dharani(192311219)

Under the Supervision of

Dr. D Sudhagaran

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

June 2026

DECLARATION

We, T Gnaneswari of the [CSE AI,CSE], SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Stuudent Skills Development management system" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 01.09.2026

Signature of the Students with Name

T Gnaneswari

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Ai-Powered Smart Requirement Analyzer For Se And Automated Srs Generation” has been carried out by T Gnaneswari,R Thrisha,G Mythili under the supervision of Dr. Radhika S ,Dr Sarala V and is submitted in partial fulfilment of the requirements for the current semester of the B.E [Computer Science and Engineering] program at Saveetha Institute of Medical and Technical Sciences, Chennai.

SIGNATURE

Dr. Rashmitha khilar

Program Director

Department of CSE

Saveetha School of Engineering

SIMATS

SIGNATURE

Dr. D Sudhagarn

Professor

Department of CSE

Saveetha School of Engineering

SIMATS

Submitted for the Project work Viva-Voce held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our Capstone Project. We are deeply thankful to our respected Founder and Chancellor, **Dr. N.M. Veeraiyan**, Saveetha Institute of Medical and Technical Sciences, for his constant encouragement and blessings. We also express our sincere thanks to our Pro-Chancellor, **Dr. Deepak Nallaswamy Veeraiyan**, and our Vice-Chancellor, **Dr. Ashwin Kumar**, for their visionary leadership and moral support during the course of this project.

We are truly grateful to our Director, **Dr. Ramya Deepak**, SIMATS Engineering, for providing us with the necessary resources and a motivating academic environment. Our special thanks to our Principal, **Dr. B. Ramesh** for granting us access to the institute's facilities and encouraging us throughout the process. We sincerely thank our Head of the Department **Dr. Rashmitha khilar** for her continuous support, valuable guidance, and constant motivation.

We are especially indebted to our guide, **Dr. D Sudhagaran** for her creative suggestions, consistent feedback, and unwavering support during each stage of the project. We also express our gratitude to the Project Coordinators, Review Panel Members (Internal and External), and the entire faculty team for their constructive feedback and valuable inputs that helped improve the quality of our work. Finally, we thank all faculty members, lab technicians, our parents, and friends for their continuous encouragement and support.

Signature With Student Name

T Gnaneswari

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Individual Contribution Statement

Student	**Specific Responsibility**	**Design & Development**	**Testing**	**Report**	**Contribution**
TGnaneswari (192372183)	Profile & Skills	Login, profile, skill module	Module testing	Introduction	25%
Madhavi (192372133)	Skill Assessment	Assessment & score calculation	Assessment testing	Implementation	25%
Ramysri (192372144)	Skill Gap & Recommendation	Gap analysis & recommendations	Result validation	Results	25%
Dharani (1923711219)	Progress & Admin	Analytics & admin dashboard	System testing	Conclusion	25%
Total					100%

ABSTRACT

The Student Skills Development Management System is a web-based application designed to support students in identifying, developing, assessing, and monitoring their technical and soft skills. In the present competitive academic and employment environment, students are expected to possess not only subject knowledge but also programming, communication, problem-solving, teamwork, leadership, and other career-oriented skills. However, conventional academic management systems mainly focus on marks, attendance, and course completion and provide limited support for continuous skill-development tracking.

The proposed system provides a centralized platform where students can create their profiles, select career goals, add existing skills, take skill assessments, monitor their performance, identify skill gaps, access learning recommendations, and track their development over time. Skills are categorized into technical and soft skills and can be represented using proficiency levels and progress percentages.

A major feature of the system is Skill Gap Analysis, which compares the student's current skill level with the required skill level for a selected career goal. Based on the identified gaps, the system recommends areas that require improvement. Students can participate in learning activities and assessments, update their progress, and unlock achievements as they develop their skills.

The system also includes an Admin/Faculty Dashboard that allows authorized faculty members to monitor students, view assessment results, analyze skill development, identify students who require improvement, and observe department-level performance through charts and reports.

The expected outcome is an accessible and user-friendly platform that improves transparency in student skill development and helps students prepare for academic, professional, and placement opportunities.

Keywords: Student Skill Development, Skill Assessment, Skill Gap Analysis, Career Development, Recommendation System, Progress Tracking, Web Application, Student Management.

Table of Content

CHAPTER 1: INTRODUCTION

1.1 Background Information

The development of appropriate technical and professional skills has become an important part of higher education. Academic marks provide information about a student's performance in a particular subject, but they may not completely represent the student's practical knowledge, technical abilities, communication skills, problem-solving ability, teamwork, leadership, and career readiness.

Students pursuing engineering and technology programs are expected to continuously develop skills according to changing industry requirements. For example, a student interested in becoming an Artificial Intelligence Engineer may require strong knowledge of Python, Machine Learning, Data Structures, SQL, mathematics, communication, and problem solving. Similarly, a student interested in becoming a Full Stack Developer may require HTML, CSS, JavaScript, databases, backend development, Git, and deployment knowledge.

Students often use multiple websites, learning platforms, certificates, assessments, and personal records to maintain information about their skills. Because this information is distributed across different platforms, monitoring overall development becomes difficult.

Faculty members also face challenges in identifying the strengths and weaknesses of individual students. Academic performance alone may not provide sufficient information about whether a student possesses the skills required for internships, placements, projects, or higher studies.

The Student Skills Development Management System is proposed as a centralized platform for maintaining and analyzing student skills. The system allows students to create profiles, select career goals, maintain technical and soft skills, complete assessments, identify skill gaps, receive recommendations, track progress, and view achievements.

The platform is designed to answer practical questions such as:

- What are my strongest skills?
- Which skills need improvement?
- What is my current proficiency level?

- Which skills are required for my career goal?
- What is the gap between my current and target skill levels?
- Which skill should I improve first?
- How has my skill performance changed over time?
- Am I becoming ready for my selected career path?

1.2 Project Objectives

The main objective of the Student Skills Development Management System is to develop an integrated web-based platform for managing, assessing, analyzing, and improving student skills.

Specific Objectives

1. To develop a secure student registration and authentication system.
2. To create and maintain individual student profiles.
3. To allow students to record technical and soft skills.
4. To classify student skills according to proficiency levels such as Beginner, Intermediate, Advanced, and Expert.
5. To conduct skill-based assessments for evaluating student knowledge.
6. To automatically calculate assessment scores and performance percentages.
7. To identify skill gaps by comparing current student skills with required skills.
8. To provide recommendations for improving weaker skills.
9. To provide learning resources for selected skills.
10. To track student skill development over time.
11. To provide graphical representations of student performance.
12. To introduce achievements and badges for encouraging continuous skill development.
13. To generate student skill-development reports.
14. To provide an Admin/Faculty Dashboard for monitoring student development.

15. To support better preparation for internships, placements, projects, and career opportunities.

1.3 Significance of the Project

The Student Skills Development Management System is significant because it extends student management beyond academic marks and attendance.

Centralized Skill Management

Students can maintain their technical and soft skills within a single platform instead of maintaining separate records.

Skill Awareness

The system helps students understand their strengths and identify skills that require improvement.

Career-Oriented Development

Students can select a career goal and compare their current skill levels with the skills expected for that career.

Continuous Assessment

Assessments provide a measurable method of evaluating student knowledge instead of relying completely on self-reported proficiency.

Skill Gap Identification

The system automatically determines the difference between current and target skill levels.

Personalized Improvement

Students receive recommendations based on their identified skill gaps.

Faculty Monitoring

Faculty members can monitor individual and overall student development through the administrative dashboard.

Data Visualization

Charts and progress indicators make skill development easier to understand.

1.4 Scope

The scope of the Student Skills Development Management System includes:

- Student registration and login.
- Faculty/Admin authentication.
- Student profile creation.
- Career goal selection.
- Technical skill management.
- Soft skill management.
- Skill proficiency tracking and Skill assessments.
- Automatic score calculation.
- Skill gap analysis and Skill recommendations.
- Learning-resource management.
- Course progress tracking.
- Student progress visualization.
- Achievement and badge management.
- Student skill reports and Admin student monitoring.
- Department-wise analytics.
- Performance visualization.

Out of Scope

The current system does not guarantee employment or placement opportunities. Skill scores generated by the application represent performance within the available assessments and should not be considered a replacement for formal professional certification.

1.5 Methodology Overview

The proposed methodology consists of several stages.

Step 1 – Student Registration

The student creates an account by entering personal and academic information.

Step 2 – Profile Creation

The system stores student information such as name, register number, department, year, semester, interests, and career goal.

Step 3 – Skill Selection

Students add their existing technical and soft skills.

Step 4 – Skill Assessment

Students complete assessments related to selected skills.

Step 5 – Score Calculation

The system calculates the number of correct answers, total score, percentage, and corresponding skill level.

Step 6 – Skill Gap Analysis

Current student performance is compared with the target skill level required for the selected career.

Step 7 – Recommendation

The system recommends skills that require further development.

Step 8 – Learning and Development

Students access learning resources and update their learning progress.

Step 9 – Faculty Monitoring

Faculty members monitor student skill performance using the Admin Dashboard.

Step 10 – Report Generation

The system generates a consolidated skill-development report.

CHAPTER 2: PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Students face several challenges when attempting to systematically develop skills for academic and professional growth.

The first challenge is the lack of centralized skill tracking. Students may learn different skills through academic courses, online learning platforms, internships, workshops, projects, and certifications. However, this information is often maintained separately.

The second challenge is difficulty in identifying weaknesses. Students may know multiple technologies but may not have a clear understanding of their actual proficiency levels.

The third challenge is career-skill mismatch. A student may have several skills but may not know whether those skills are sufficient for a selected career goal.

The fourth challenge is limited continuous assessment. Students often evaluate themselves informally without using structured assessments.

The fifth challenge is lack of progress monitoring. Without historical performance data, it becomes difficult to understand whether a student's skill level is actually improving.

The sixth challenge is faculty monitoring. Faculty members may have difficulty individually monitoring the skill-development progress of a large number of students.

The seventh challenge is fragmented learning resources. Students frequently use different platforms to learn individual skills without a structured development path.

The proposed Student Skills Development Management System addresses these problems through centralized skill management, assessments, skill-gap analysis, recommendations, learning tracking, analytics, and faculty monitoring.

2.2 Evidence of the Problem

Modern employment opportunities increasingly require candidates to demonstrate practical and professional skills in addition to academic qualifications. Technical roles may require programming, database management, problem solving, cloud technologies, artificial intelligence, data analysis, web development, or cybersecurity skills.

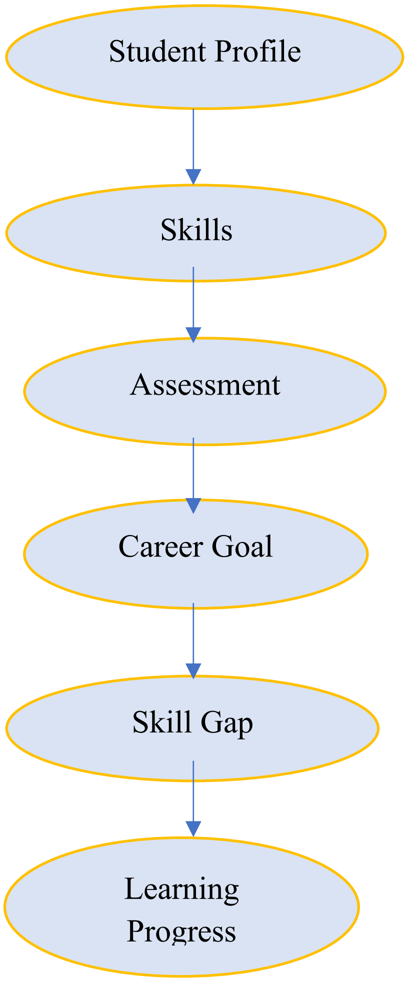
Employers also consider communication, teamwork, adaptability, leadership, presentation ability, and critical thinking.

Traditional academic systems primarily maintain information such as:

- Student details
- Attendance
- Subject marks
- Examination results
- Course registration

These systems may provide limited information about the continuous development of career-oriented skills.

Therefore, there is a requirement for a system that connects:



Existing Approaches and Limitations

Existing Approach	Limitation
Traditional student portals	Mainly focus on academic information
Manual skill records	Difficult to maintain and analyze
Self-assessment	May not accurately measure knowledge
Individual learning platforms	Skill information remains distributed
Academic marks	Do not completely represent practical skills
Manual faculty monitoring	Difficult for large numbers of students
Generic learning recommendations	May not consider individual skill gaps

2.3 Stakeholders

Stakeholder	Interest / Requirement
Students	Identify and improve skills
Faculty	Monitor student development
Mentors	Guide students based on weaknesses
Placement Team	Understand student career readiness
Department	Analyze overall skill performance
Administrators	Manage users and system information
Recruiters	Potentially understand candidate capabilities
Institution	Improve student skill-development activities

2.4 Supporting Data / Research

The system requires structured information related to students, skills, assessments, career requirements, courses, and progress.

The student profile can contain:

- Student name
- Register number
- Department

- Year
- Semester
- Email
- Interests
- Career goal

Skill information can contain:

- Skill name
- Skill category
- Current level
- Target level
- Progress percentage
- Assessment score
- Status

Career-skill information can contain:

- Career name
- Required skills
- Target proficiency
- Skill priority

Assessment information can contain:

- Skill
- Question
- Options
- Correct answer
- Difficulty

CHAPTER 3: SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The project follows an iterative software-development methodology similar to the phased approach used in the sample Capstone report.

Phase 1 – Requirement Analysis

Student and faculty requirements are identified.

Phase 2 – Data Design

Student, skill, assessment, course, achievement, and progress data structures are designed.

Phase 3 – System Design

System architecture, modules, database collections, user interfaces, and workflows are designed.

Phase 4 – Frontend Development

The user interface is developed for registration, login, dashboard, skills, assessments, recommendations, progress, and administration.

Phase 5 – Backend Development

Backend APIs are implemented for authentication, skill management, assessment processing, progress tracking, and administration.

Phase 6 – Database Development

Database collections are created for users, skills, assessments, results, courses, achievements, and progress.

Phase 7 – Skill Analysis Development

Algorithms for score calculation and skill-gap analysis are implemented.

Phase 8 – Integration

Frontend, backend, database, and analytics components are integrated.

Phase 9 – Testing

Functional, validation, authentication, usability, and performance testing are performed.

Phase 10 – Evaluation

The final application is evaluated based on functionality, calculation accuracy, usability, and response time.

3.2 Tools and Technologies Used

Category	Technology
Programming Language	Python
Frontend	HTML, CSS, JavaScript
UI Framework	Bootstrap
Backend	FastAPI
Database	MongoDB
Visualization	Chart.js
API Architecture	REST API
Server	Uvicorn
Development Environment	VS Code / Gravity
Browser	Google Chrome
Version Control	Git / GitHub

Python and FastAPI are suitable for developing backend services because they provide a structured approach for implementing APIs and application logic.

MongoDB provides flexible document-based storage for student profiles, skills, assessments, and activity information.

Chart.js is used to represent student skill development and assessment performance graphically.

3.3 Solution Overview

The Student Skills Development Management System consists of several major components.

1. User Authentication Module

Provides secure registration, login, logout, and role-based access.

2. Student Profile Module

Stores:

- Name
- Register Number
- Email
- Department
- Year
- Semester
- Interests
- Career Goal

3. Skill Management Module

Allows students to add, modify, update, and remove technical and soft skills.

4. Assessment Module

Conducts skill-based quizzes and automatically calculates results.

5. Skill Gap Analysis Module

Compares current student skill levels with target levels.

6. Recommendation Module

Suggests areas for improvement according to identified gaps.

7. Learning Module

Provides learning resources and allows students to track completion.

8. Progress Module

Displays skill and assessment development through charts.

9. Achievement Module

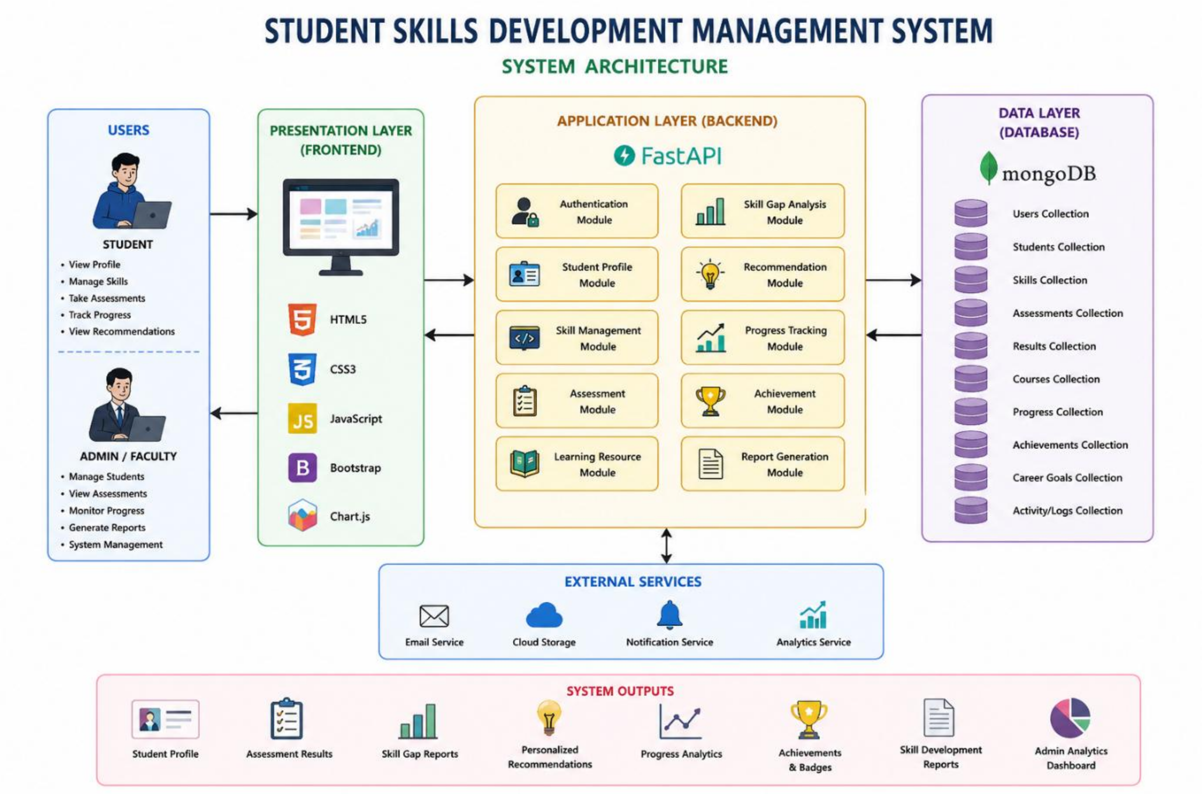
Provides badges for reaching milestones.

10. Admin/Faculty Module

Allows authorized faculty members to monitor student development.

3.4 System Architecture

The system architecture follows the flow:



The architecture separates the presentation, application, processing, and database components, making the system easier to maintain and extend.

3.5 Functional Modules

Module 1: User Authentication

Students and faculty members securely access the application using authorized credentials.

Module 2: Student Profile

Stores personal, academic, career, and interest information.

Module 3: Skill Management

Students maintain technical and soft skills.

Module 4: Skill Assessment

The system evaluates students using skill-specific questions.

Module 5: Score Calculation

Assessment responses are evaluated and converted into percentage scores.

Module 6: Skill Gap Analysis

The system determines differences between current and required skill levels.

Module 7: Recommendation

Skills with larger gaps receive higher improvement priority.

Module 8: Learning Resources

Students can access learning content for targeted skills.

Module 9: Progress Tracking

Skill development is represented through progress bars and charts.

Module 10: Achievements

The system rewards predefined milestones with badges.

Module 11: Admin Dashboard

Faculty members can monitor students, assessments, and progress.

Module 12: Report Generation

A consolidated student skill-development report is generated.

3.6 Algorithm and Methodology

Skill Assessment Algorithm

```
START

Get Student ID

Select Skill

Load Assessment Questions

score = 0

FOR each question

   Get student answer

   IF student answer = correct answer

      score = score + 1

   END IF

END FOR

percentage = (score / total questions) × 100

IF percentage < 40

   level = Beginner

ELSE IF percentage < 60

   level = Intermediate

ELSE IF percentage < 80

   level = Advanced

ELSE

   level = Expert

END IF

Save result
```

Display score, percentage and level

STOP

Skill Gap Analysis Algorithm

```
START

Get Student ID

Get Career Goal

Retrieve current skills

Retrieve required skills for career

FOR every required skil

    current = Student Skill Percentage

    required = Required Skill Percentage

    gap = required - current

    IF gap > 0

        Status = Needs Improvement

    ELSE

        gap = 0

        Status = Requirement Achieved

    END IF

    Store Result

END FOR

Sort skills according to largest gap

Generate recommendations

Display Skill Gap Analysis

STOP
```

Formula

Skill Gap = Required Skill Level − Current Skill Level

Example:

Required Python = 90%

Current Python = 70%

Skill Gap = 90 − 70 = 20%

Therefore, the student requires approximately 20% improvement in Python relative to the configured target.

3.7 Engineering Standards Applied

Standard / Principle	Application
ISO/IEC 25010	Software quality evaluation
ISO/IEC 27001 Principles	Information security
IEEE 29148	Requirements engineering
IEEE 1016	Software design documentation
Software Testing Principles	Test documentation
OWASP Principles	Web application security

The sample report similarly considers software quality, requirements engineering and web-security principles when discussing engineering standards.

ISO/IEC 25010

The application can be evaluated based on:

- Functional suitability
- Performance efficiency
- Usability
- Reliability
- Security
- Maintainability

- Compatibility
- Portability

Security Considerations

Authentication and authorization should prevent unauthorized access to student information.

Input validation helps prevent invalid data from entering the application.

Passwords should not be stored directly in plain text.

Role-based access prevents students from accessing administrative functions.

3.8 Solution Justification

The proposed solution is justified because conventional academic systems primarily provide academic information and may not provide detailed continuous skill-development analysis.

A marks-management application can show examination performance but may not identify career-specific skill gaps.An online learning platform may provide courses but may not maintain institution-level student development information.A self-assessment system can measure individual skills but may not provide continuous progress tracking.

The Student Skills Development Management System integrates these components:

Student Profile → Skills → Assessment → Skill Gap → Recommendation → Learning → Progress → Analytics

This provides a more comprehensive platform for supporting student skill development.

CHAPTER 4: RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The effectiveness of the Student Skills Development Management System can be evaluated using several parameters.

Functional Accuracy

The application should correctly process registration, authentication, skill management, assessment, and progress operations.

Assessment Accuracy

Assessment answers should be correctly evaluated.

Skill Gap Accuracy

The difference between target and current skill levels should be calculated correctly.

Recommendation Relevance

Recommendations should correspond to the student's actual identified skill gaps.

Data Accuracy

Student profile, skill, assessment, and progress information should be correctly stored and retrieved.

Usability

Students should be able to navigate the system and understand their skill-development information.

Performance

Pages and API operations should execute within an acceptable response time.

4.2 Output Parameters

Parameter	Expected Outcome
Registration	Successful account creation
Authentication	Secure user access
Student Profile	Correct information storage
Skill Management	Correct skill CRUD operations
Assessment	Accurate score generation
Skill Level	Correct proficiency classification
Skill Gap	Correct difference calculation
Recommendations	Relevant improvement suggestions
Progress	Correct graphical representation
Achievement	Badge unlocked at required milestone
Admin Analytics	Correct student summaries
Report	Accurate consolidated information

4.3 Test Cases and Expected/Actual Results

Test Case	Expected Result	Actual Result	Status
Student Registration	Account created	Account created	PASS
Valid Login	Dashboard displayed	Dashboard displayed	PASS
Invalid Login	Error message	Error displayed	PASS
Add Skill	Skill saved	Skill saved	PASS
Edit Skill	Skill updated	Skill updated	PASS
Delete Skill	Skill removed	Skill removed	PASS
Assessment	Score generated	Score generated	PASS
Skill Gap	Gap correctly calculated	Correct result	PASS
Recommendation	Weak skills recommended	Recommendations shown	PASS
Course Progress	Progress updated	Progress updated	PASS
Achievement	Badge unlocked	Badge displayed	PASS
Admin Dashboard	Student information displayed	Information displayed	PASS

Important: Change the "Actual Result" and "PASS" values according to your real testing before submission.

Test Success Rate

Test Success Rate (%) = Passed Test Cases / Total Test Cases × 100

If all 12 tests pass: Test Success Rate = 12 / 12 × 100 = 100%

4.4 Challenges Encountered

Challenge 1 – Maintaining Different Skill Types

Students can possess several technical and soft skills with different proficiency levels.

Solution: Skills are stored using structured categories, proficiency levels, and percentages.

Challenge 2 – Skill-Level Measurement

Self-reported skill levels may not always represent actual knowledge.

Solution: Skill assessments are included to provide additional measurable evidence.

Challenge 3 – Career-Skill Mapping

Different careers require different combinations of skills.

Solution: Career goals are mapped to predefined target skills and proficiency levels.

Challenge 4 – Recommendation Relevance

Generic recommendations may not be useful to every student.

Solution: Recommendations are generated according to identified skill gaps.

Challenge 5 – Data Visualization

Large amounts of student performance information can be difficult to understand as plain text.

Solution: Charts and progress indicators are used.

Challenge 6 – Access Control

Students should not be allowed to access administrative functionality.

Solution: Role-based authorization is implemented.

4.5 Possible Improvements

Future versions of the system can include:

1. Machine-learning-based skill recommendations.
2. AI-based career recommendations.
3. Resume skill extraction.
4. Automatic resume analysis.
5. Placement-readiness prediction.
6. Company-specific skill-gap analysis.
7. Integration with online learning platforms.
8. Certificate verification.
9. Internship recommendation.
10. Project recommendation according to student skills.
11. AI-based mock interviews.
12. Coding assessment integration.
13. Mobile application support.
14. Faculty mentoring and appointment features.
15. Email and mobile notifications.
16. Advanced department analytics.
17. Peer skill comparison.
18. Personalized learning paths.

4.6 Recommendations

The following recommendations are proposed for future deployment:

- Regularly update career-specific skill requirements.
- Maintain a larger assessment question bank.
- Validate assessment questions before publishing.
- Provide transparent skill scoring.
- Implement strong authentication and access control.
- Regularly back up student information.
- Conduct usability testing with actual students.
- Collect student feedback.
- Provide diverse learning resources.
- Continuously improve skill recommendations.

4.7 SDG RELEVANCE

SDG 4 – Quality Education

The project primarily supports Sustainable Development Goal 4: Quality Education.

The system encourages continuous learning by helping students identify existing skills, detect weaknesses, complete assessments, and monitor development.

It can support students in developing technical and professional competencies in addition to academic knowledge.

SDG 8 – Decent Work and Economic Growth

The system also relates to SDG 8: Decent Work and Economic Growth.

Career-oriented skill development can improve students' preparation for internships, placements, entrepreneurship, and employment opportunities.

By identifying skill gaps before students enter the employment market, the platform can support better career preparation.

CHAPTER 5: REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

Academic Knowledge

The project provided an opportunity to apply concepts from computer science, database management, software engineering, web development, data visualization, algorithms, and application security.

The project demonstrated how theoretical concepts can be converted into a practical student-development platform.

Technical Skills

During the development of the project, technical knowledge was developed in:

- Python programming
- FastAPI
- HTML
- CSS
- JavaScript
- Bootstrap
- MongoDB
- REST API development
- Authentication
- Database operations
- Chart.js
- Data visualization
- Software testing

- Git/GitHub

Problem Solving and Critical Thinking

The project involved multiple interconnected problems.

For example, a student may have a high proficiency in one technical skill but a significant gap in another skill required for the selected career.

The system therefore requires analysis of multiple skill values rather than simply storing student information.

Developing the project improved the ability to divide a large problem into smaller functional modules and integrate them into a complete application.

5.2 Challenges Encountered and Overcome

One challenge was designing a flexible structure capable of maintaining different skills for different students.

Another challenge was designing a method for converting assessment performance into understandable skill levels.

Skill-gap analysis also required the current student level and target career level to be represented consistently.

Frontend and backend integration required careful management of API requests, responses, validation, and database operations.

Authentication and authorization required separation between student and administrative access.

These challenges improved debugging, logical thinking, application design, and testing skills.

5.3 Application of Engineering

Software-engineering principles provided a structured approach to the development of the project.Requirement analysis was used to identify student and faculty requirements.Modular design was used to divide the system into authentication, profile, skill, assessment, recommendation, progress, and administrative components.Database-design concepts were applied when creating structures for users, skills, assessments, and results.

5.4 Insights into the Industry

The project provided an understanding that modern software development involves more than programming.

A complete software application requires:

- Requirement analysis
- User interface design
- Backend development
- Database management
- Security
- Testing
- Performance evaluation
- Deployment
- Documentation
- Maintenance
- Continuous improvement

5.5 Conclusion of Personal Development

The Student Skills Development Management System contributed to both technical and professional learning.

Technically, the project improved understanding of frontend development, backend APIs, databases, authentication, algorithms, analytics, and testing.

The development process also strengthened problem-solving, communication, teamwork, documentation, debugging, and time-management skills.

The project demonstrated how software-engineering knowledge can be applied to solve a practical educational problem.

CHAPTER 6: CONCLUSION

The Student Skills Development Management System was developed as a centralized solution for identifying, assessing, managing, and improving student skills.

The project addresses the limitations of traditional academic management systems by extending student monitoring beyond marks and attendance. The proposed platform maintains technical and soft skills, performs assessments, calculates proficiency levels, identifies skill gaps, provides recommendations, monitors learning progress, and displays development through analytics.

One of the major advantages of the system is its integrated approach. Instead of maintaining student skills, assessments, learning activities, and progress separately, these components are connected within a single platform.

The Skill Gap Analysis module helps students understand the difference between their current capabilities and career-oriented target requirements. Recommendations then provide guidance regarding which skills require additional attention.

The Admin/Faculty Dashboard further improves the usefulness of the system by allowing faculty members to monitor student performance and development.

The project demonstrates the practical application of web development, database management, software engineering, algorithms, data visualization, testing, and security principles to an educational problem.

Future versions of the system can incorporate machine learning, AI-based career guidance, resume analysis, placement-readiness prediction, automated certificate verification, learning-platform integration, and mobile application support.

Overall, the Student Skills Development Management System demonstrates how digital technology can support continuous learning, career preparation, skill awareness, and student development.

REFERENCES

For the final submission, include only references that you actually used in the project. Suitable categories are:

1. FastAPI – Official Documentation.
2. MongoDB – Official Documentation.
3. Python – Official Documentation.
4. Bootstrap – Official Documentation.
5. Chart.js – Official Documentation.
6. OWASP Foundation – Web Application Security Guidelines.
7. ISO/IEC 25010 – Systems and Software Quality Models.
8. ISO/IEC/IEEE 29148 – Requirements Engineering.
9. Research papers related to student skill assessment and career recommendation systems.
10. Relevant educational skill-development datasets, if any dataset is actually used.

APPENDICES

Appendix A – Sample Student Input

Input	Example
Name	Student A
Register Number	192372183
Department	CSE-AI
Year	IV
Semester	VII
Career Goal	AI Engineer
Python	70%
Machine Learning	55%
SQL	65%
Communication	80%

Appendix B – Sample System Output

Career Goal: AI Engineer

Skill	Current	Required	Gap	Result
Python	70%	90%	20%	Improve
Machine Learning	55%	85%	30%	Improve
SQL	65%	70%	5%	Improve
Communication	80%	70%	0%	Achieved

Recommendation

Priority 1: Improve Machine Learning – Gap 30%

Priority 2: Improve Python – Gap 20%

Priority 3: Improve SQL – Gap 5%

Strength: Communication – Required level achieved.

Appendix C – Execution Screenshots

After completing the website, paste your actual screenshots in this order:

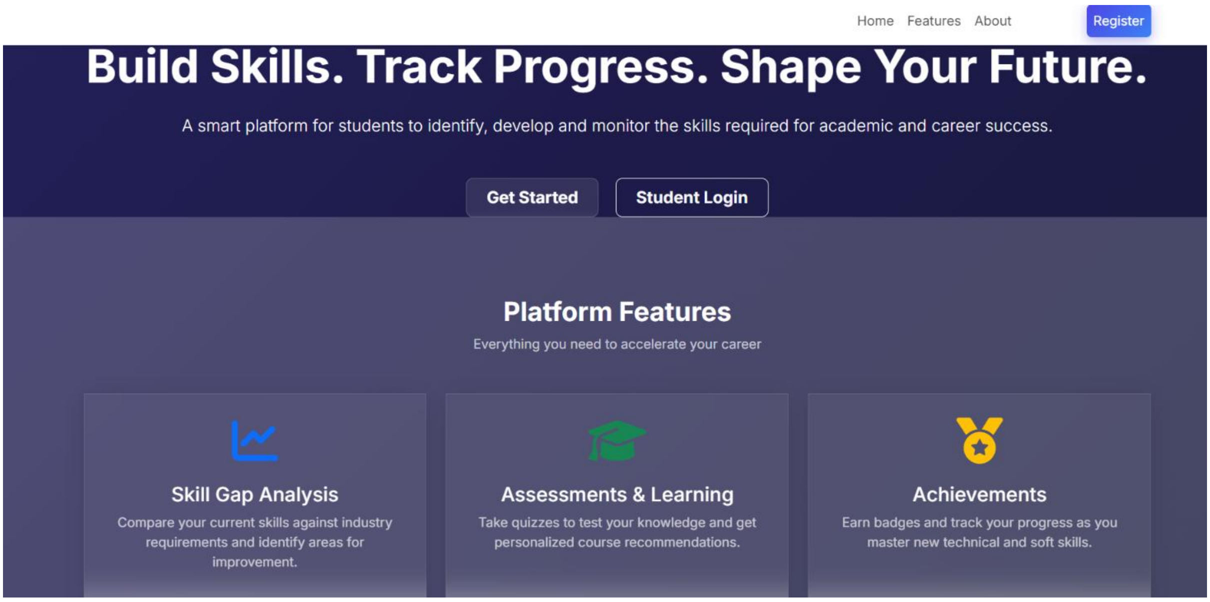


Figure 1 – Home Page

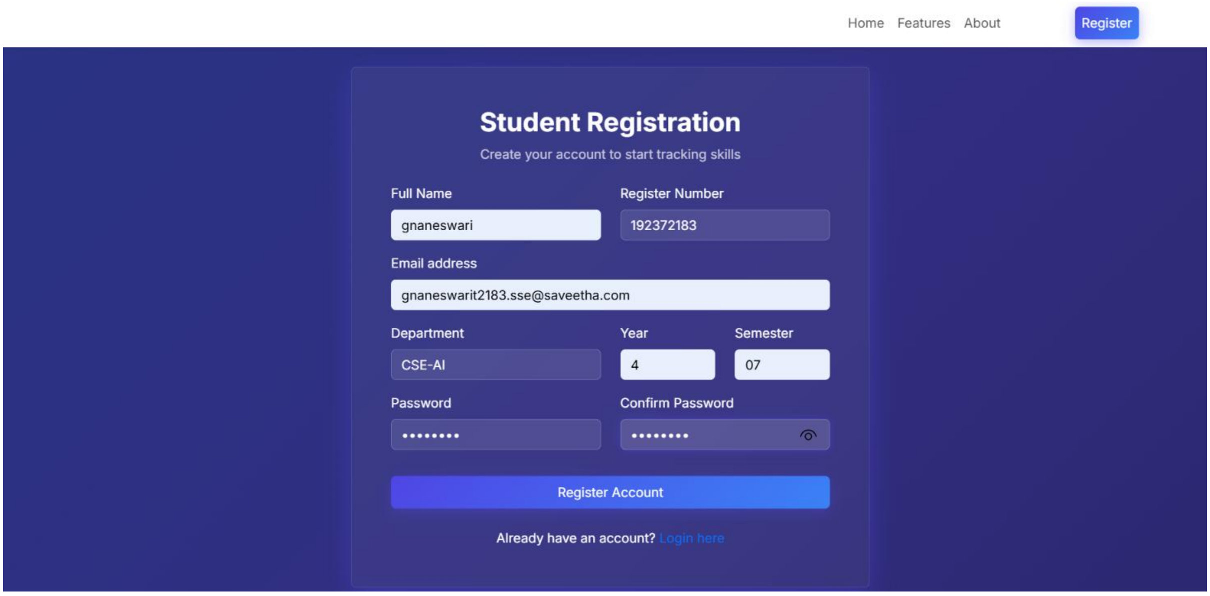


Figure 2 – Student Registration Page

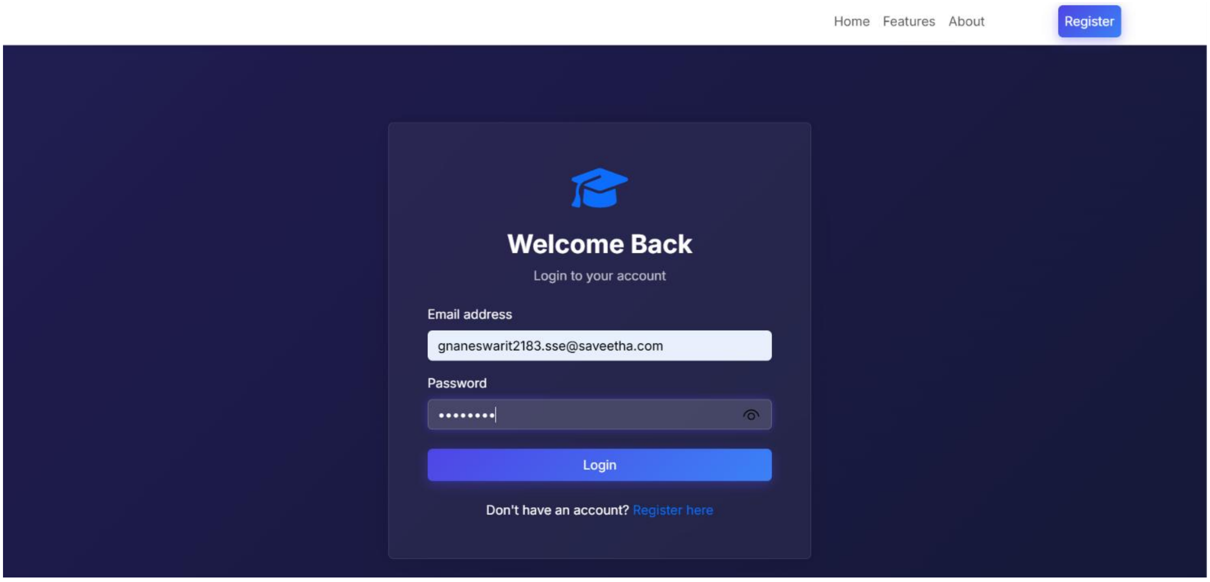


Figure 3 – Student Login Page

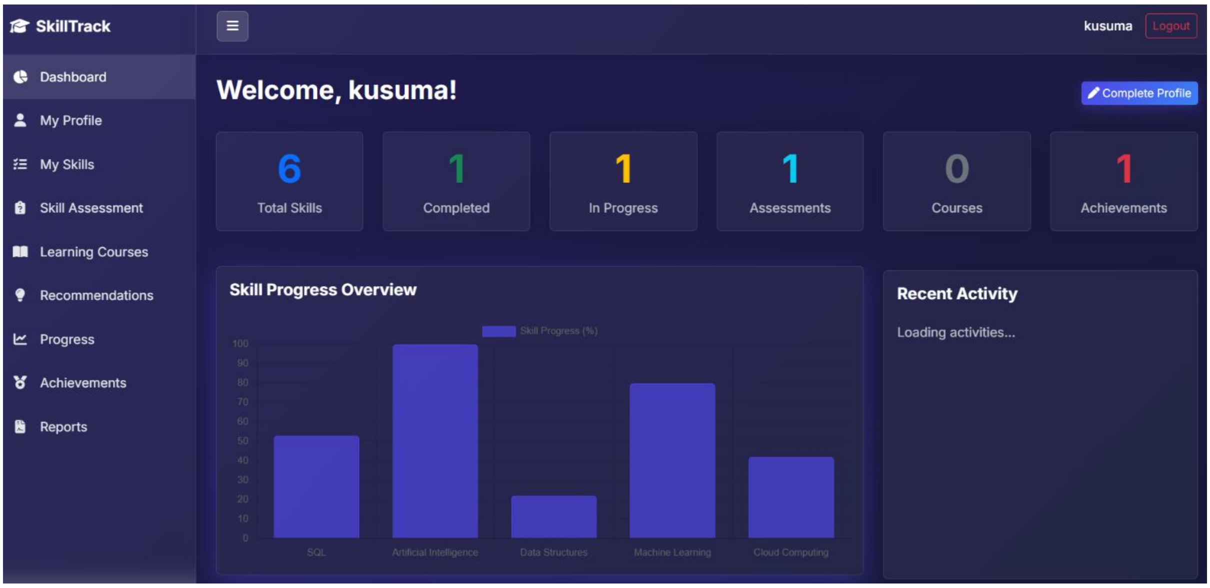


Figure 4 – Student Dashboard

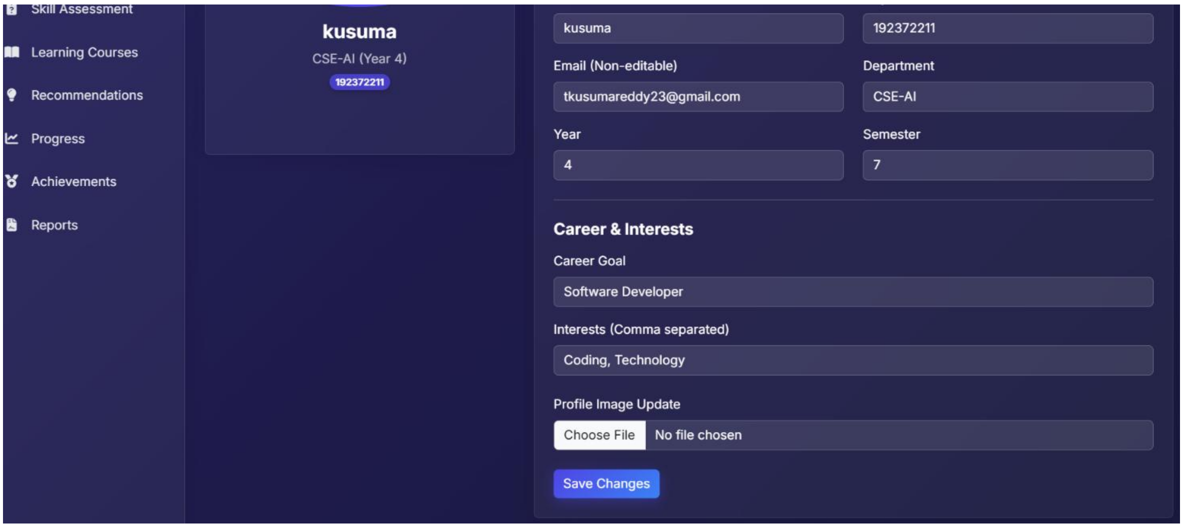


Figure 5 – Student Profile

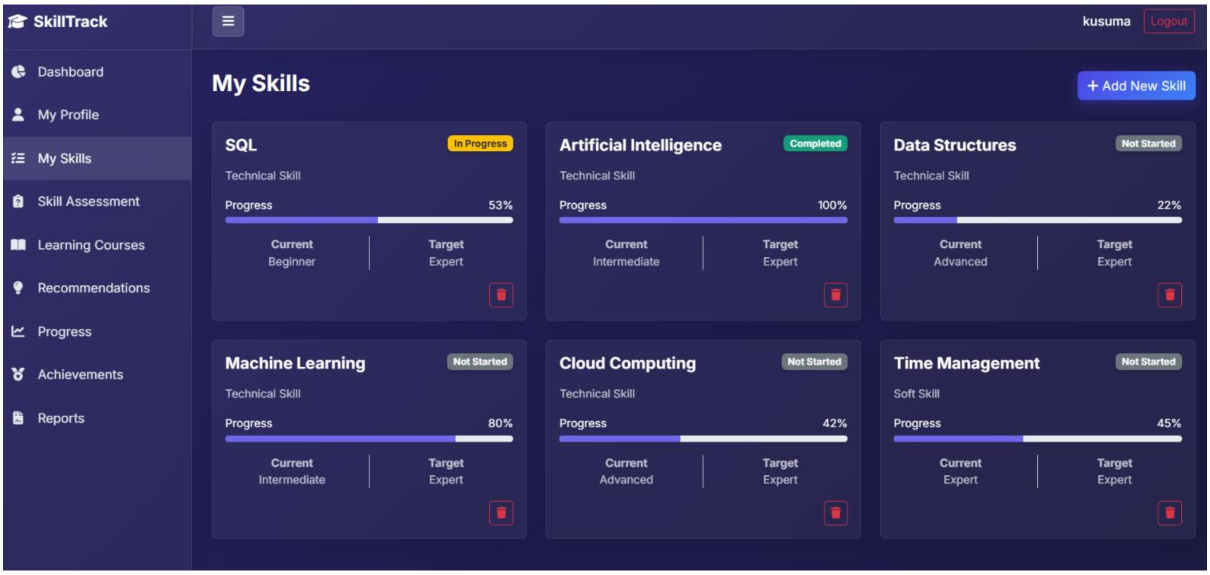


Figure 6 – My Skills

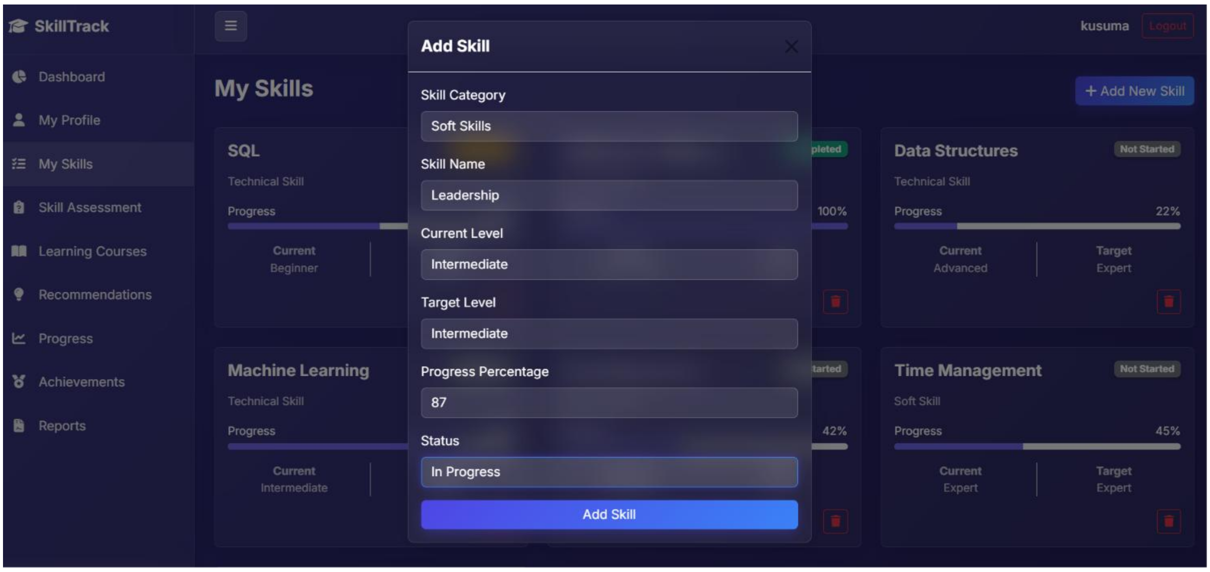


Figure 7 – Add/Update Skill

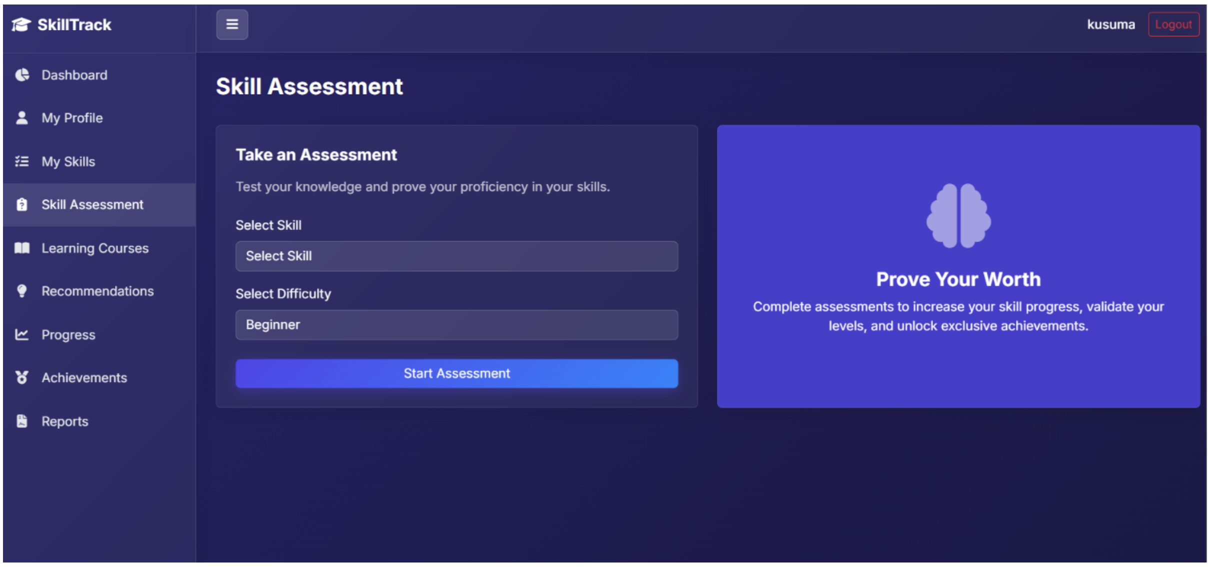


Figure 8 – Skill Assessment

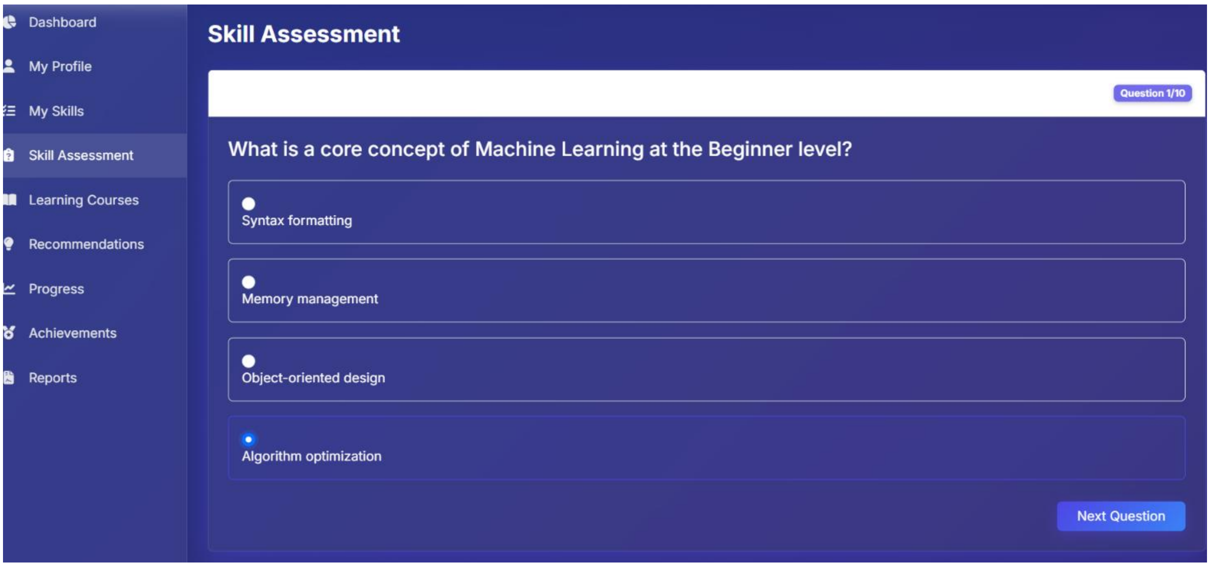


Figure 9 – Assessment Result

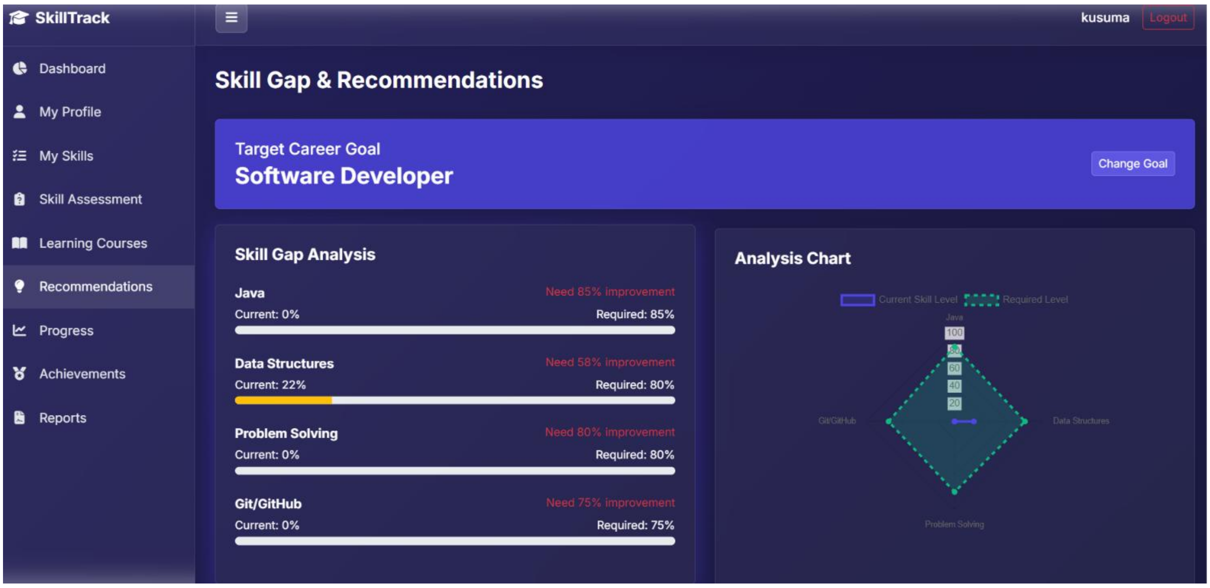


Figure 10 – Skill Gap Analysis

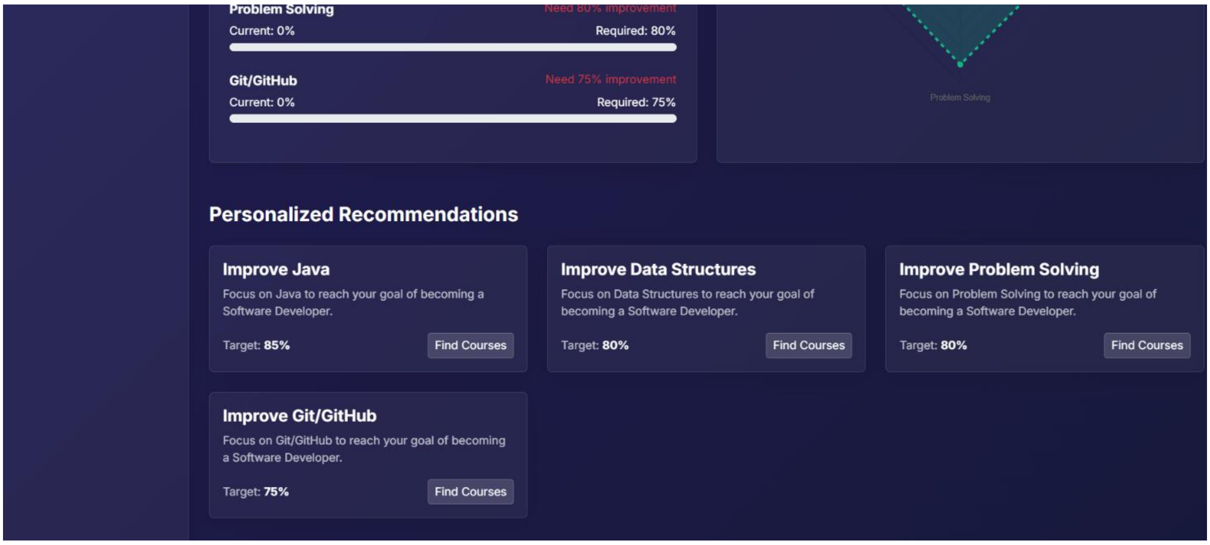


Figure 11 – Personalized Recommendations

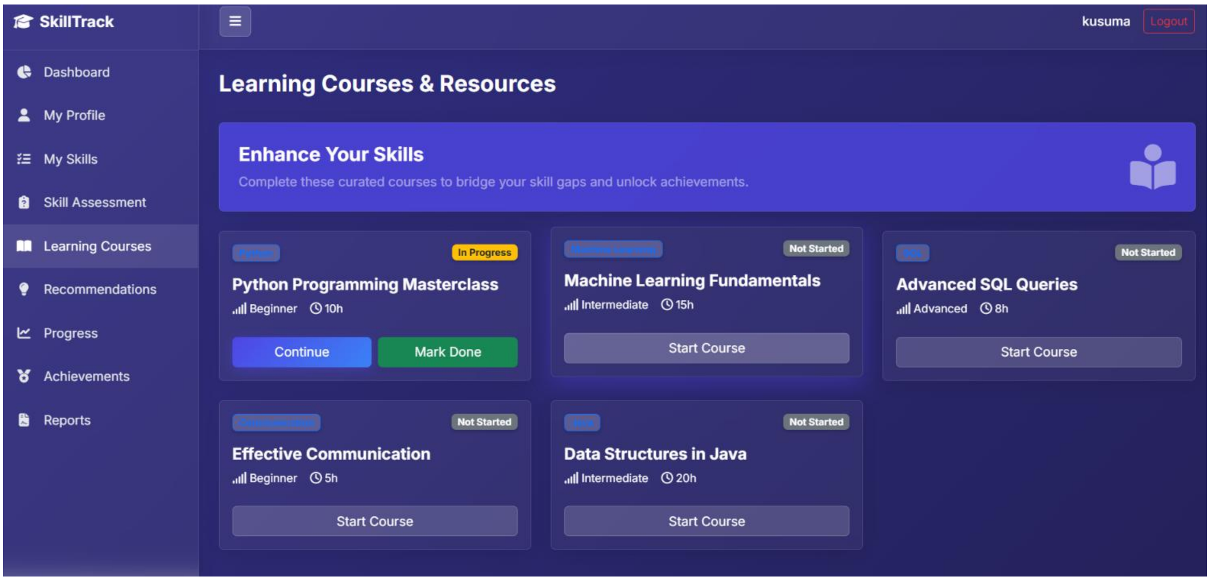


Figure 12 – Learning Courses

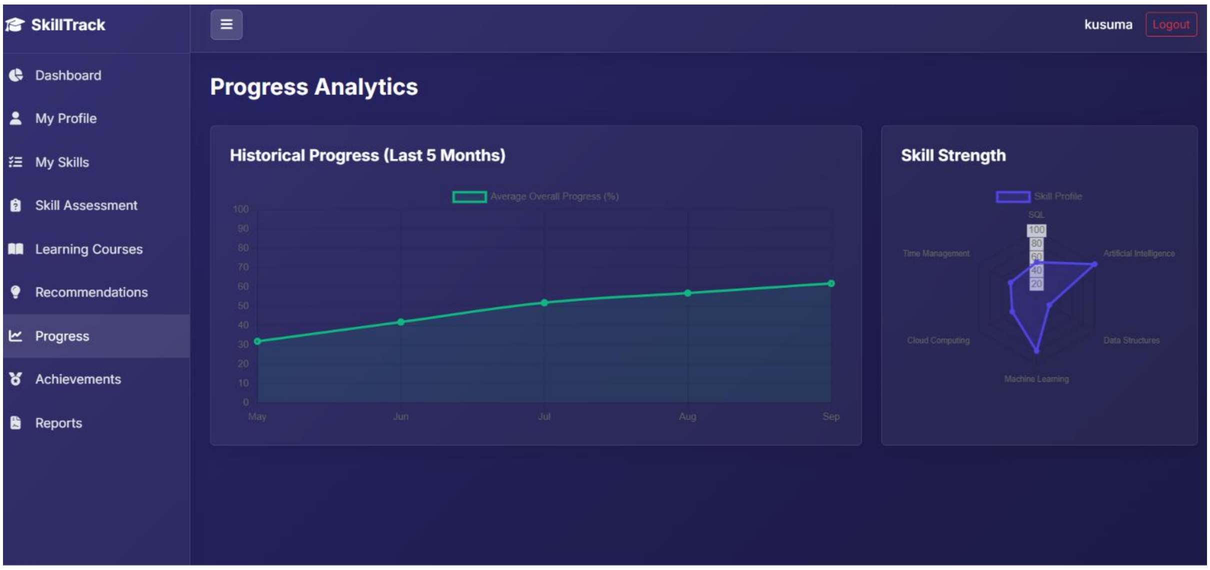


Figure 13 – Progress Analytics

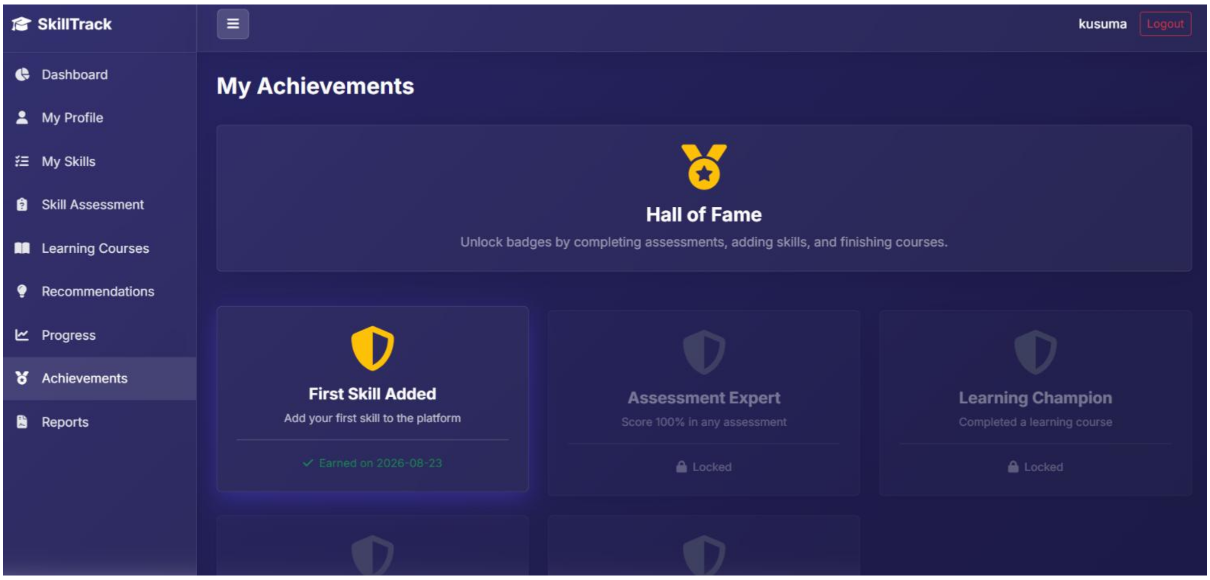


Figure 14 – Achievements

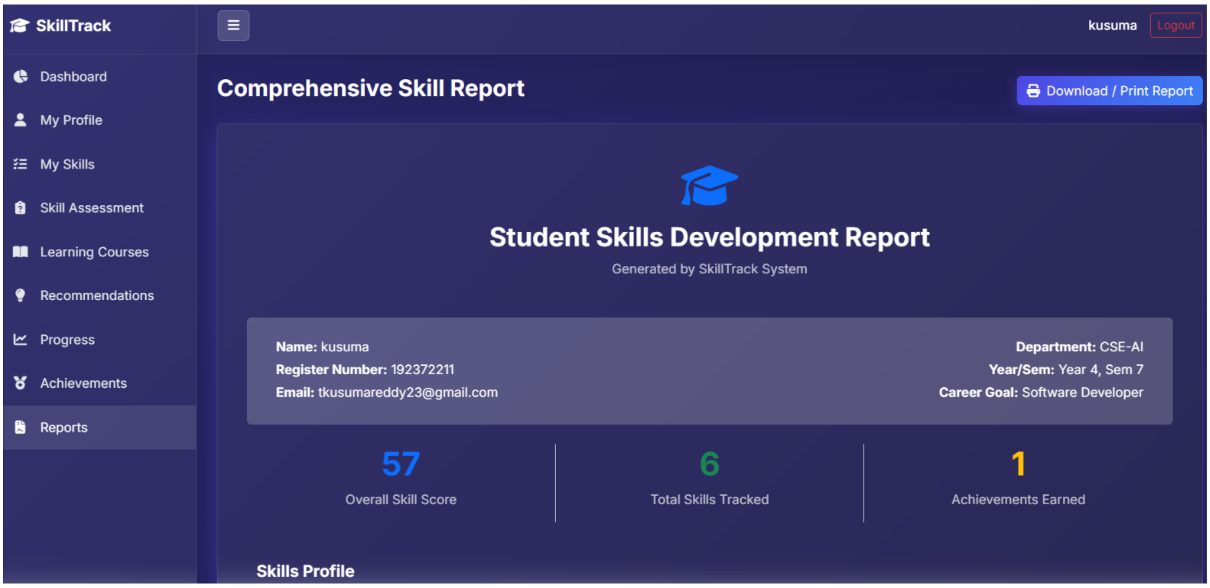


Figure 15 – Student Skill Report

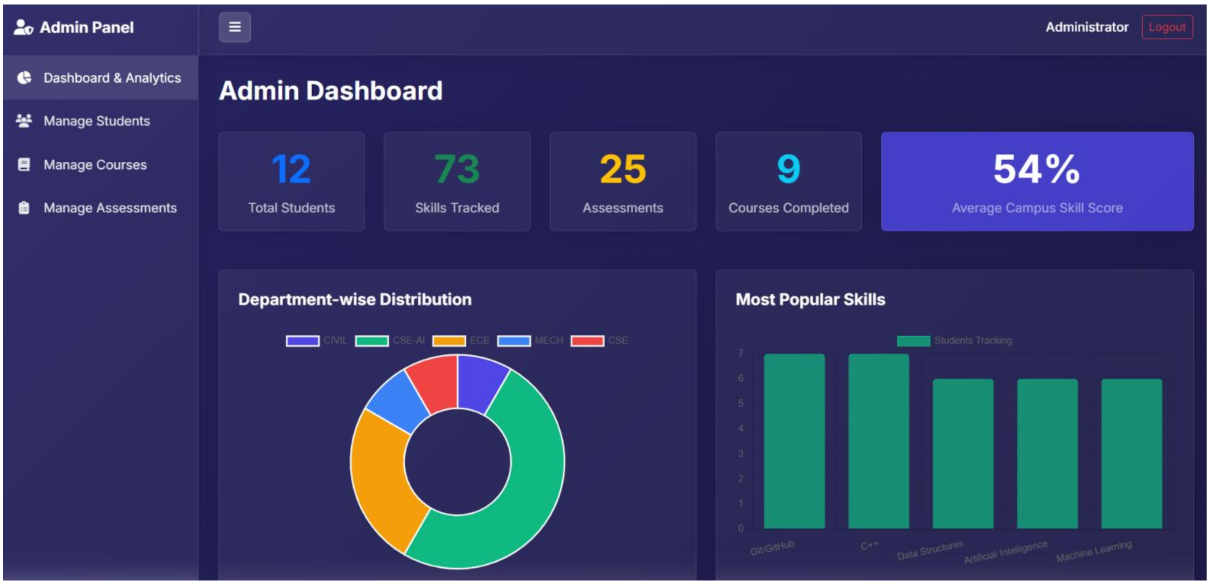
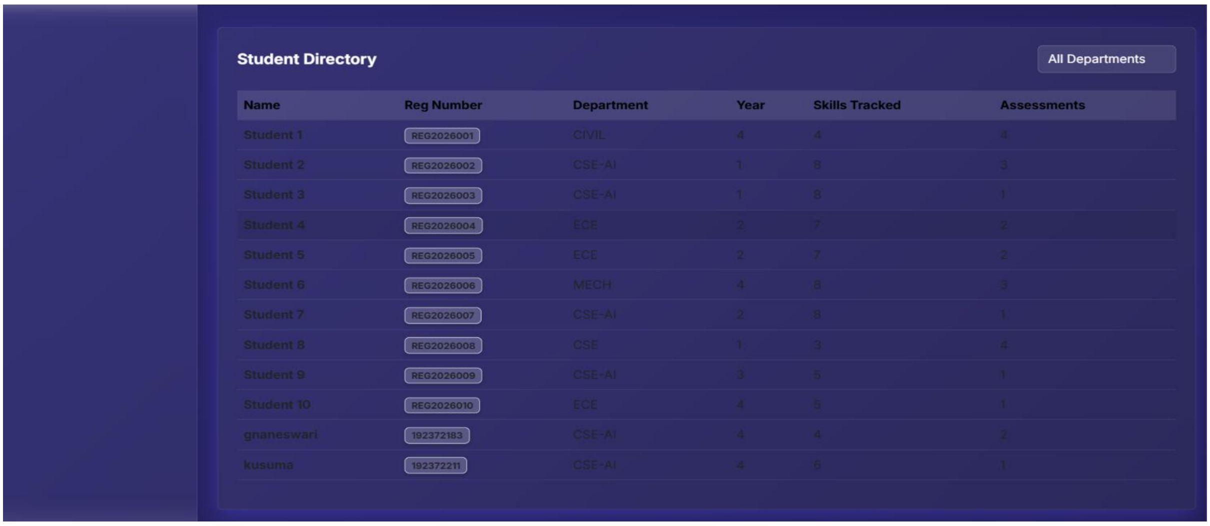


Figure 16 – Admin Dashboard



Student Directory

All Departments

Name	Reg Number	Department	Year	Skills Tracked	Assessments
Student 1	REG2026001	CIVIL	4	4	4
Student 2	REG2026002	CSE-AI	1	8	3
Student 3	REG2026003	CSE-AI	1	8	1
Student 4	REG2026004	ECE	2	7	2
Student 5	REG2026005	ECE	2	7	2
Student 6	REG2026006	MECH	4	8	3
Student 7	REG2026007	CSE-AI	2	8	1
Student 8	REG2026008	CSE	1	3	4
Student 9	REG2026009	CSE-AI	3	5	1
Student 10	REG2026010	ECE	4	5	1
gnaneswari	192372183	CSE-AI	4	4	2
kusuma	192372211	CSE-AI	4	6	1

Figure 17 – Student Management

Figure 18 – Admin Analytics